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# MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

*(A constituent unit of MAHE, Manipal)*

## VII SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

## END SEMESTER EXAMINATIONS, NOVEMBER 2024

## INTRODUCTION TO ELECTRIC VEHICLES [ELE4084]

**Time: 3 Hours** **Date: 20 NOV 2024** **Max. Marks: 50**

**Instructions to Candidates:**

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

**1A.** Discuss the need for electric vehicle penetration & its adoption, challenges, and possible solutions with regard to the Indian Scenario. **04**

**1B.** Derive the mathematical equation to determine the road-load power by considering different forces acting on the vehicle. **03**

**1C.** The Tesla Model S is traveling down a $-8°$ slope at 120 km/h. The mass of the vehicle is 2155 kg and vehicle parameters are A=177.2 N; B= 1.445 $N/ms^{-1}$; C=0.35 $N/_{m^2s^{-2}}$ respectively. Assuming calm conditions, how much regenerative power is available to brake the vehicle while maintaining a constant speed? **03**

**2A.** Explain the different power flow control modes of a typical parallel hybrid system with the help of block diagrams. **03**

**2B.** Analyze the differences between micro, mild, and full hybrid technologies in transportation, highlighting their advantages, limitations, and applications. **03**

**2C.** The Nissan Leaf electric vehicle has a battery energy of 90 kWh. The vehicle parameters are A=133.3 N, B=0.7094 $N/ms^{-1}$ , C=0.491 $N/(ms^{-1})^2$ respectively. Furthermore, the efficiency of the power train from the battery to the transmission is 85%.

i. Estimate the range of the above electric vehicle at 120 km/h.

ii. Determine the reduction in range for the above BEV if the vehicle has a continuous heating,ventilation, and air conditioning (HVAC) load of 6 kW. **04**

**3A.** Determine the beginning-of-life kilowatt-hour storage required in a BEV battery pack based on the following requirements: 6 years of operation, an average of 55 km of driving per day over the 365 days of the year, daily charging, and an average battery output energy per kilometer is 180 Wh/km. Assume the battery pack cycle lifetime index is 3 and the number of charge\discharge cycles for 100% depth of discharge is 1000. Assume two parallel battery strings with 96 Li-ion cells per string, with a total number of cells are 192, and a nominal voltage of 3.75 V per cell. Assume that the capacity of the battery pack will reduce to 80% at the end of life.

i. Estimate the ampere-hours per cell.

ii. Evaluate the vehicle ranges at the beginning of life (BOL) and end of life **04**

(EOL).

**3B.** Explain the basic principle of supercapacitor-based energy storage systems in hybrid electric vehicles. **03**

**3C.** Explain the dynamic characteristics of a fuel cell with its equivalent electrical model. **03**

**4A.** A fully loaded fuel cell-based tractor has the following parameters: mass of 36,280 kg, drag coefficient of 0.7, vehicle cross-section area of 10 $m^2$, and coefficient of rolling resistance of 0.008. The nominal efficiency of the powertrain and transmission is 85 %, and the auxiliary load is 2 kW. Let the density of the air is 1.2 Kg/$m^3$. The overall mass of the fuel cell system (including the fuel cell, balance of plant, storage tanks, and mechanical bracketing) is 400 kg plus 80 kg per 5 kg of stored hydrogen.

i. Determine the miles per gallon equivalent (mpge), and the overall mass of the fuel cell system if the vehicle is to travel at a constant speed of 64 km/h for two work shifts of 8 hours total, or 512 km. The fuel cell plant efficiency is 50% for this operating condition.

ii. Evaluate the mass of the battery which would be required if the specific energy is 0.15 kWh/kg.

**04**

**4B.** Identify and discuss the key features of a PMSM motor and outline the method used to control its speed. **03**

**4C.** Illustrate with a diagram and explain the operation of a four-quadrant chopper in controlling the speed of a DC motor. **03**

**5A.** Analyze and contrast the key features of various battery charging methods. **03**

**5B.** Explain the role of the power electronic converter employed in battery electric vehicles to capture the energy generated during regenerative braking of the vehicle with a neat circuit diagram. **04**

**5C.** Discuss the implementation issues of energy management strategies for an HEV. **03**